Interpersonal Liking (H5)

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```r
library(ggeffects)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.1     v stringr   1.5.2
## v ggplot2   4.0.0     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(ggplot2)
library(lme4)
```

```
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
```

```r
library(lmerTest)
```

```
##
## Attaching package: 'lmerTest'
##
## The following object is masked from 'package:lme4':
##
##     lmer
##
## The following object is masked from 'package:stats':
##
##     step
```

```r
library(patchwork)
```

```r
# Load Interleaf data
df <- read.csv("../../Data/df_interleaf_measures.csv")
```

```r
# Mean-center Likability (per participant, to reduce multicollinearity)
df <- df %>%
  group_by(participant_ID) %>%
  mutate(likability_centered = likability_rating - mean(likability_rating, na.rm = TRUE)) %>%
  ungroup()
```

```r
# Sum Contrasts: Code HAI as -0.5 and HH as 0.5
# What this does: Recode condition order as -0.5 and 0.5
# prevents the use of dummy contrasts,
# Interpretation: Slope represents raw difference between conditions
# Intercept each is the grand mean of the dv in the condition

df$shared_condition_order <- as.factor(as.character(df$shared_condition_order))
contrasts(df$shared_condition_order) <- c(-0.5, 0.5)
```

```r
# Print Contrast Matrix
contrasts(df$shared_condition_order)
```

```
##             [,1]
## with_ai     -0.5
## without_ai   0.5
```

```r
# Check that everything is there and working
attributes(df$shared_condition_order)
```

```
## $levels
## [1] "with_ai"    "without_ai"
##
## $class
## [1] "factor"
##
## $contrasts
##             [,1]
## with_ai     -0.5
## without_ai   0.5
```

```r
# H5: Liking, Flow
likability_flow_model <- lmer(
  flow_rating ~ shared_condition_order * likability_centered +
                chat_number + shared_prompt_order +
                (1 + shared_condition_order | participant_ID) +
                (1 | dyad_ID),
  data = df
)
summary(likability_flow_model)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: flow_rating ~ shared_condition_order * likability_centered +
##     chat_number + shared_prompt_order + (1 + shared_condition_order |
##     participant_ID) + (1 | dyad_ID)
##    Data: df
##
## REML criterion at convergence: 2833.8
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.4892 -0.5001  0.0672  0.5791  2.2069
##
## Random effects:
##  Groups         Name                    Variance Std.Dev. Corr
##  participant_ID (Intercept)             272.43   16.506
##                 shared_condition_order1 143.43   11.976   0.06
##  dyad_ID        (Intercept)              50.65    7.117
##  Residual                               172.50   13.134
## Number of obs: 336, groups:  participant_ID, 42; dyad_ID, 21
##
## Fixed effects:
##                                             Estimate Std. Error        df
## (Intercept)                                 47.57712    3.74990  42.84158
## shared_condition_order1                     -9.60407    2.37432  42.13932
## likability_centered                          0.69916    0.04794 256.85111
## chat_number                                  1.31900    0.34489 276.58107
## shared_prompt_order                          0.19459    0.65460 249.93070
## shared_condition_order1:likability_centered  0.16156    0.10462 271.68748
##                                            t value Pr(>|t|)
## (Intercept)                                 12.688 4.19e-16 ***
## shared_condition_order1                     -4.045 0.000218 ***
## likability_centered                         14.583  < 2e-16 ***
## chat_number                                  3.824 0.000162 ***
## shared_prompt_order                          0.297 0.766516
## shared_condition_order1:likability_centered  1.544 0.123678
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) shr__1 lkblt_ cht_nm shrd__
## shrd_cndt_1 -0.002
## lkblty_cntr  0.164 -0.173
## chat_number -0.365  0.054 -0.255
## shrd_prmpt_ -0.408  0.021 -0.132 -0.086
## shrd_cn_1:_ -0.031  0.002 -0.021 -0.194  0.120
```

```
# H5: Liking, Social Presence
likability_social_presence_model <- lmer(
  social_presence_rating ~ shared_condition_order * likability_centered +
              chat_number + shared_prompt_order +
              (1 + shared_condition_order | participant_ID) +
              (1 | dyad_ID),
  data = df
```

```r
)
summary(likability_social_presence_model)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## social_presence_rating ~ shared_condition_order * likability_centered +
##     chat_number + shared_prompt_order + (1 + shared_condition_order |
##     participant_ID) + (1 | dyad_ID)
##    Data: df
##
## REML criterion at convergence: 2866
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.1681 -0.4605  0.0428  0.4507  3.5018
##
## Random effects:
##  Groups         Name                    Variance Std.Dev. Corr
##  participant_ID (Intercept)             265.97   16.309
##                 shared_condition_order1 367.16   19.162   -0.35
##  dyad_ID        (Intercept)              46.43    6.814
##  Residual                               178.64   13.366
## Number of obs: 336, groups:  participant_ID, 42; dyad_ID, 21
##
## Fixed effects:
##                                             Estimate Std. Error        df
## (Intercept)                                 62.75266    3.74045  50.16210
## shared_condition_order1                     -8.32333    3.32658  42.12148
## likability_centered                          0.73029    0.05188 286.84273
## chat_number                                  0.25376    0.35586 267.77021
## shared_prompt_order                          0.18220    0.66728 248.28403
## shared_condition_order1:likability_centered   0.37651    0.10622 272.81821
##                                            t value Pr(>|t|)
## (Intercept)                                 16.777  < 2e-16 ***
## shared_condition_order1                     -2.502 0.016315 *
## likability_centered                         14.077  < 2e-16 ***
## chat_number                                  0.713 0.476419
## shared_prompt_order                          0.273 0.785047
## shared_condition_order1:likability_centered   3.545 0.000462 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) shr__1 lkblt_ cht_nm shrd__
## shrd_cndt_1 -0.238
## lkblty_cntr  0.181 -0.133
## chat_number -0.380  0.043 -0.266
## shrd_prmpt_ -0.418  0.018 -0.143 -0.083
## shrd_cn_1:_ -0.036  0.006 -0.054 -0.186  0.125
```

```r
# H5: Liking, Affect
likability_panas_model <- lmer(
  panas_rating ~ shared_condition_order * likability_centered +
                 chat_number + shared_prompt_order +
                 (1 + shared_condition_order | participant_ID) +
                 (1 | dyad_ID),
  data = df
)
summary(likability_panas_model)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: panas_rating ~ shared_condition_order * likability_centered +
##     chat_number + shared_prompt_order + (1 + shared_condition_order |
##     participant_ID) + (1 | dyad_ID)
##    Data: df
##
## REML criterion at convergence: 2685.2
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.4266 -0.4519 -0.0092  0.4771  3.5786
##
## Random effects:
##  Groups         Name                    Variance Std.Dev. Corr
##  participant_ID (Intercept)             145.21   12.050
##                 shared_condition_order1  31.10    5.577   -0.48
##  dyad_ID        (Intercept)              92.81    9.634
##  Residual                               118.56   10.888
## Number of obs: 336, groups:  participant_ID, 42; dyad_ID, 21
##
## Fixed effects:
##                                             Estimate Std. Error        df
## (Intercept)                                 61.11827    3.36674  38.34463
## shared_condition_order1                     -2.15016    1.49911  42.75469
## likability_centered                          0.29262    0.03607 187.73243
## chat_number                                  0.31940    0.27817 287.69996
## shared_prompt_order                          0.28362    0.54203 251.21307
## shared_condition_order1:likability_centered   0.01425    0.08658 273.23944
##                                            t value Pr(>|t|)
## (Intercept)                                 18.154  < 2e-16 ***
## shared_condition_order1                     -1.434    0.159
## likability_centered                          8.113 6.29e-14 ***
## chat_number                                  1.148    0.252
## shared_prompt_order                          0.523    0.601
## shared_condition_order1:likability_centered   0.165    0.869
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) shr__1 lkblt_ cht_nm shrd__
## shrd_cndt_1 -0.183
## lkblty_cntr  0.136 -0.206
```

```
## chat_number -0.325  0.058 -0.223
## shrd_prmpt_ -0.375  0.024 -0.125 -0.091
## shrd_cn_1:_ -0.036  0.009 -0.057 -0.184  0.123
```

```
# Plotting
#Colors
teal <- "#0397b2"
damped_light_blue <-"#b6e2ff"
```

```
# Set colors
colors <- c(
  "with_ai"    = teal,
  "without_ai" = damped_light_blue
)
```

```
# Global theme, applies to all plots
theme_set(
  theme_minimal(base_size = 20) +
    theme(
      axis.text        = element_text(size = 18),
      axis.title       = element_text(size = 19, face = "bold"),
      legend.text      = element_text(size = 17),
      legend.title     = element_text(size = 18),
      strip.text       = element_text(size = 18, face = "bold"),
      plot.title       = element_text(size = 22, face = "bold", hjust = 0.5)
    )
)
```

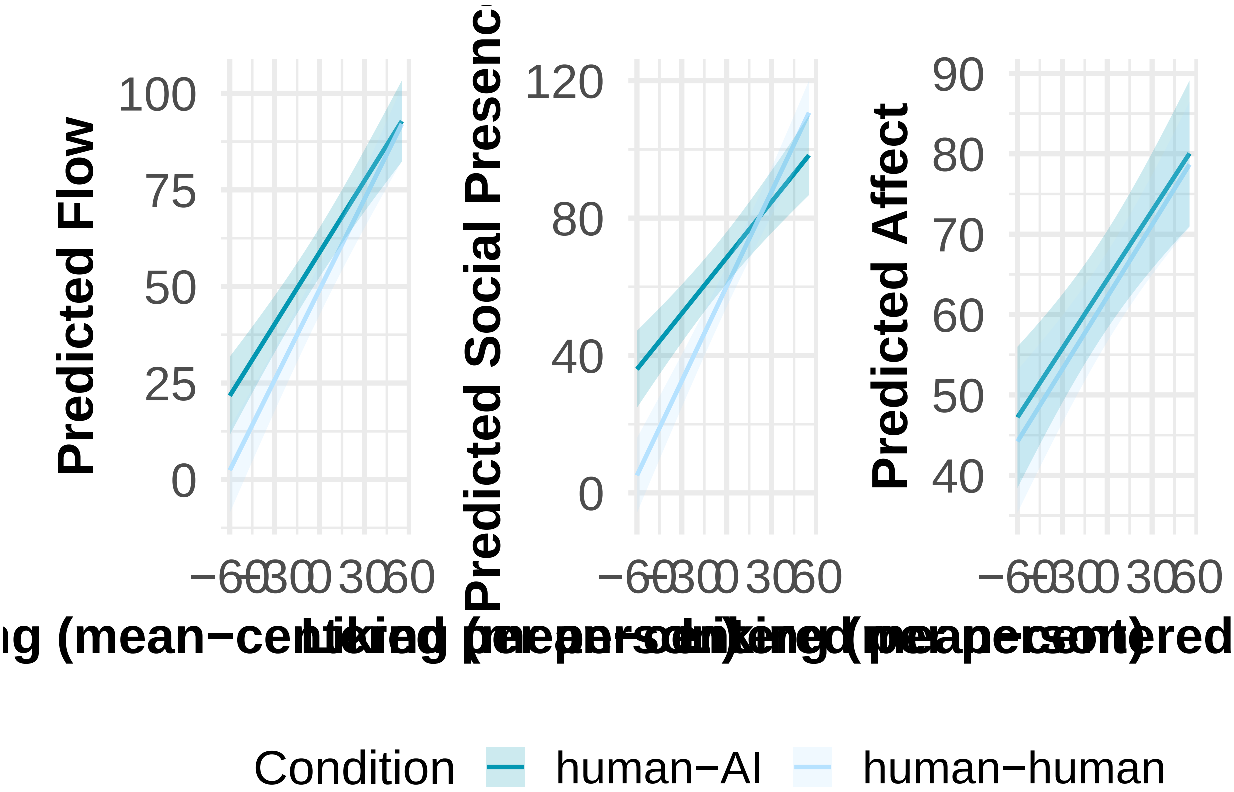
```
# Generate predictions for each model
pred_flow <- ggpredict(likability_flow_model,
                       terms = c("likability_centered", "shared_condition_order"))

pred_social <- ggpredict(likability_social_presence_model,
                         terms = c("likability_centered", "shared_condition_order"))

pred_panas <- ggpredict(likability_panas_model,
                        terms = c("likability_centered", "shared_condition_order"))

# Helper function to build each plot consistently
plot_interaction <- function(pred_data, y_label) {
  ggplot(pred_data, aes(x = x, y = predicted, color = group, fill = group)) +
    geom_line(size = 0.8) +
    geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.2, color = NA) +
    scale_color_manual(values = colors, labels = c("human-AI", "human-human")) +
    scale_fill_manual(values = colors, labels = c("human-AI", "human-human")) +
    labs(
      x = "Liking (mean-centered per person)",
      y = y_label,
      color = "Condition",
      fill = "Condition"
    ) +
    theme(
      axis.text  = element_text(size = 18),
```

```r
      axis.title = element_text(face = "bold"),
      legend.position = "none",  # remove from individual plots
      strip.text = element_text(face = "bold")
    )
}

# Build individual plots
p_flow   <- plot_interaction(pred_flow,   "Predicted Flow")
```

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```

```r
p_social <- plot_interaction(pred_social, "Predicted Social Presence")
p_panas  <- plot_interaction(pred_panas,  "Predicted Affect")

# Combine in one row, shared legend at bottom
interaction_plot <- (p_flow | p_social | p_panas) +
  plot_layout(guides = "collect") &
  theme(legend.position = "bottom")
```

```r
# Plot is optimized for the pdf output, that is produced by ggsave, not for the
# knitted PDF of the R Script!
interaction_plot
```




```r
ggsave(
  filename = "Likability_Interaction_Plots.pdf",
  plot = interaction_plot,
  width = 16,
  height = 6,
  dpi = 300,
  device = pdf
)
```